

Working with Large Datasets in R

Please access code here:

<https://columbia-data.club/3pqedb>

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Please access slides using this tinyurl:

<https://columbia-data.club/kcjbes>

Please download OpenFEMA

NFIRS datasets 2024

Moacir created a short download link for the NFIRS 2024, so you can use either the original link above or the link below for download

<https://columbia-data.club/lb8h6d>

outline

1. Large datasets in R setting
2. A few (possible) solutions
3. A few R packages



In R and RStudio, "large data" typically refers to datasets that exceed the available RAM (random access memory) of the computer running R.

- A general rule of thumb suggests needing 2-3 times the RAM of your data size for comfortable processing.
- Data exceeding 50% of available RAM are essentially unusable.
 - We consider a data set large if it exceeds 20% of the RAM on a given machine and massive if it exceeds 50%

TODAY

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- Image source: easy.peasy.ai



There are no one-fits-for-all solutions

If you are reading my slides later, solutions can be totally different from today (Oct. 9, 2025), due to hardware and software improvements

2

A few (possible) solutions on an individual's laptop or desktop

Two real world data examples



Seattle Public Library Checkouts Open Data

- Data subsetting
- Data partitioning
- Various data formats are provided



OpenFEMA guide to working with large datasets

- Download only what you need.
- To refresh data, attempt to perform updates rather than a full re-download.
- Pre-process (filter and aggregate) prior to analysis using more standard tools.
- Use tools more appropriate for working with large amounts of data—including programming languages.
- Seek additional resources from the web regarding data science, data wrangling, and data analysis tools used to work with large amounts of data.



[Moacir's Python session on Oct. 23](#) will cover more details about Polars, so I won't spend too much time today to talk about it.

I will discuss a few R packages (version number > 1.0.0)

3

You might (or must) have other options

Comparisons

1

data.table

- fread()
- Ideal for medium size CSV files within memory
- Not ideal for directly handling JSON and parquet files
- Can also consider use vroom package and compare benchmarks

2

arrow

- read_delim_arrow()
- write_parquet()
- Ideal for most of txt files including CSVs
- data size can be larger than memory

3

dbplyr
vs.
dtplyr

- dbplyr is dplyr backend packages (syntax compatibility)
- dbplyr is based on SQL, so if you don't know SQL, there is a learning curve
- dtplyr is data.table backend and slower than data.table

4

duckplyr

- Ideal for common data formats such as json, delimited files and parquet files
- can handle large data sets that do not fit into RAM
- full dplyr compatibility

5

polars

- The R packages now are not matured enough in my opinion
- If you prefer to use polars, use Python for now (Oct. 2025) rather than R



Hands-on demo in • R/Rstudio

Using [OpenFEMA NFIRS datasets 2024](#)



Thank you!

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Ask us anything!

Please find other data club sessions' info [here](#)

Any questions related to today's session please contact

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